Introduction to VLSI Testing and Manufacturing

Nguyen-Van Thanh-Loc

Email: locnvt@hcmute.edu.vn

Outline

1. Evolution of IC Technology and Testing Challenges

- Moore's Law and VLSI Growth
- Why Testing is Essential?
- Roles of VLSI Testing

2. VLSI Development Process

- Basic Testing Approach
- VLSI Development Process
- Manufacturing Testing and Yield Improvement
- Design Verification Flow

3. Yield and Manufacturing Quality

- Yield and Reject Rate
- Yield Models
- Electronic System Manufacturing Process
- System-Level Operation and Reliability

4. Exercises

- Yield
- Good Dies
- Die Cost
- IC Cost
- Defect Level
- Cost Trade-off

Evolution of IC Technology and Testing Challenges

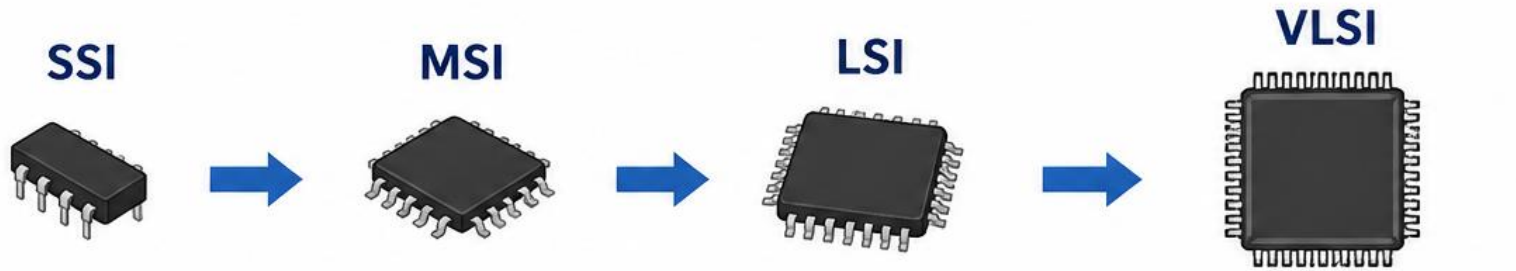
- ICs evolved from SSI to VLSI with exponentially **increasing transistor counts**.
- Higher integration leads to higher **testing complexity**.
- Modern VLSI introduces many new **testing challenges**.

Moore's Law and VLSI Growth

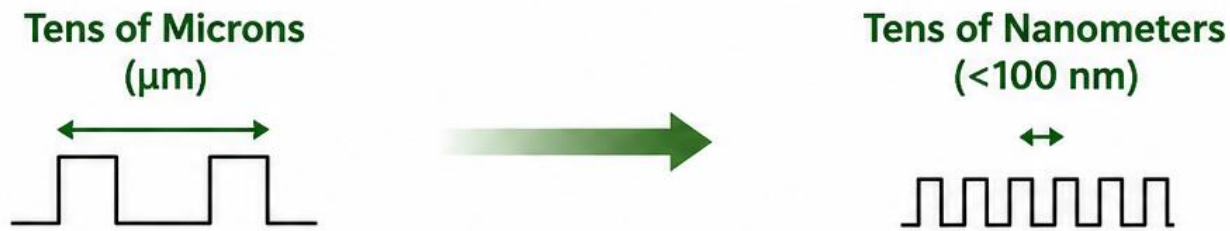
- IC density doubles every ~18 months (**Moore's Law**).
- VLSI has evolved from 100K transistors to millions and hundreds of millions of transistors.
- Technology scaling (**smaller feature size**) enables **higher integration** and **higher operating speed**.

TECHNOLOGY SCALING TREND

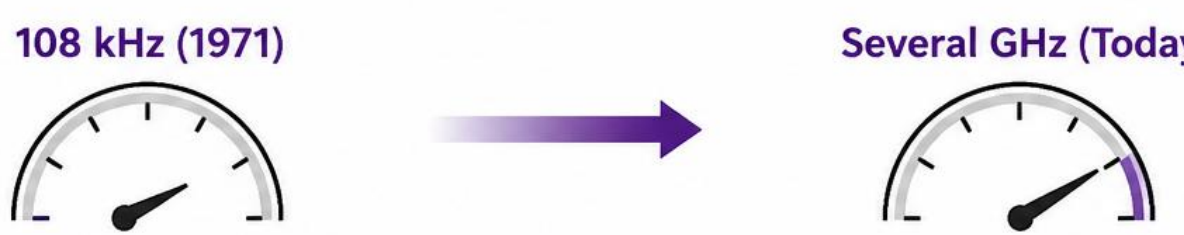
MORE TRANSISTORS



SMALLER FEATURE SIZE



HIGHER PERFORMANCE





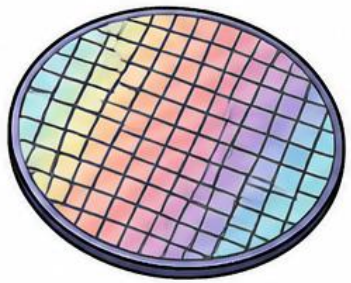
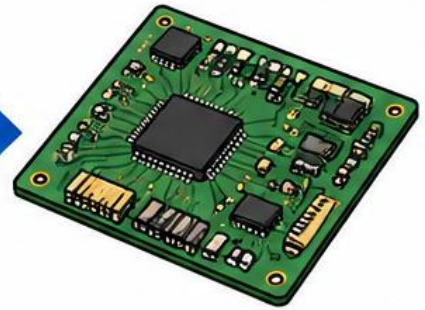
CONCLUSION → Technology scaling drives IC complexity and performance growth.

Why Testing is Essential?

- **Smaller feature sizes increase defect sensitivity**
  - As feature size decreases (<100 nm), even tiny manufacturing defects can cause faulty transistors or interconnects.
- **Manufacturing defects are unavoidable**
  - A single faulty transistor or wire can cause the entire chip to fail, and some defective chips are inevitable in any manufacturing process.
- **Early testing reduces cost**
  - Testing is required at multiple stages (IC → PCB → System), and the cost of finding a defect increases by roughly **10×** at each successive stage (*Rule of Ten*).

Rule of Ten

Cost of finding a defect increases by 10× at each subsequent stage

1. Device (Wafer)	2. Board (PCB)	3. System	4. Field (Operation)
			
Test individual die on wafer	Test assembled circuit board	Test complete system	Failure found by customer in field
1×	10×	100×	1000×
$	$$	$$$	$$$$

×10 ×10 ×10

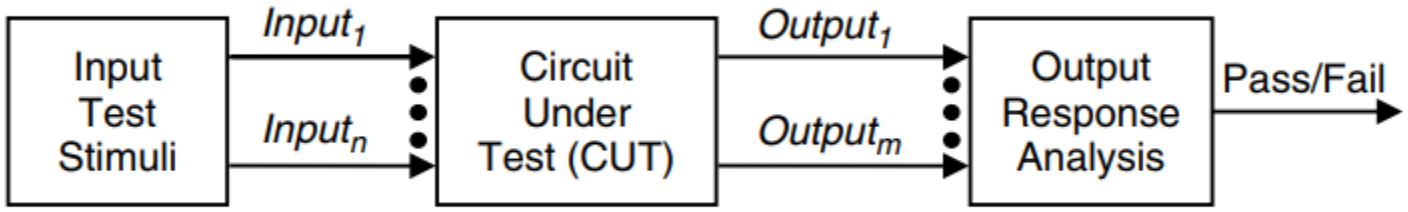
Detect earlier = Save cost!

Roles of VLSI Testing

- Testing Coverage
  - IC Testing
  - PCB Testing
  - System Testing
  - In-Field Testing

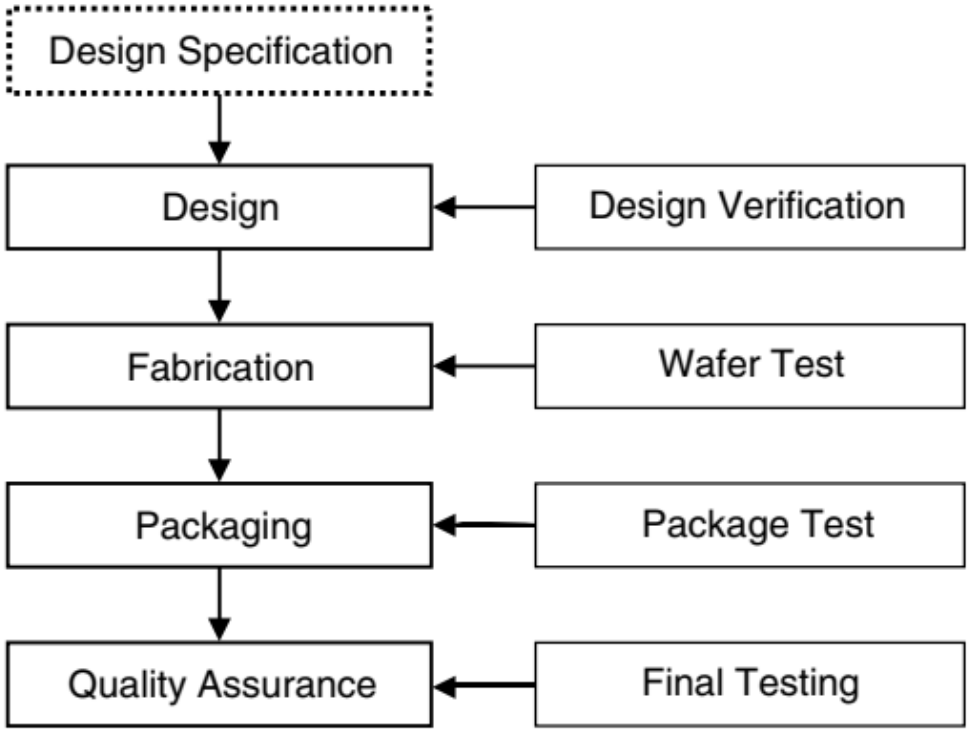
Basic Testing Approach

- Apply a set of **test stimuli** to the inputs of the **Circuit Under Test (CUT)**.
- Analyze the **output responses**.
- Correct outputs → **Pass**; Incorrect outputs → **Fail**.



■ FIGURE 1.1

VLSI Development Process (1/2)



■ FIGURE 1.2

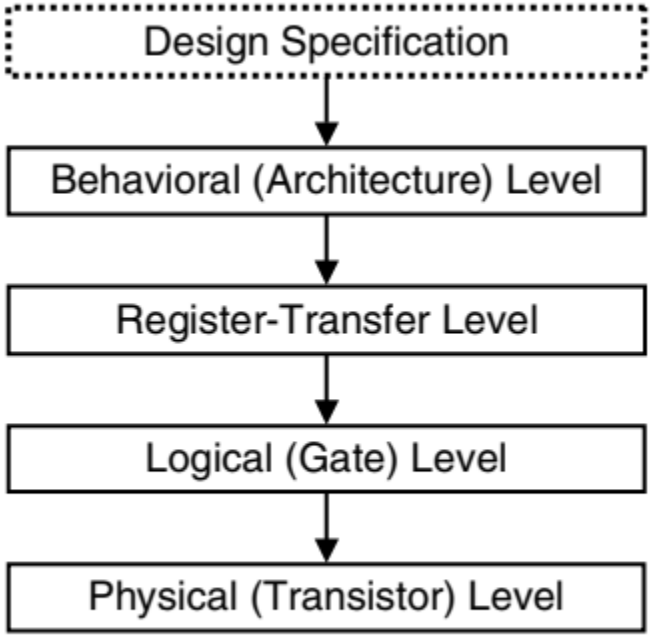
VLSI Development Process (2/2)

- Some form of **testing is involved** at each stage of the **VLSI development process**.
- Designers **synthesize a circuit** based on the design specification.
- **Design verification** ensures the design will perform the required functions **before manufactured**.

Manufacturing Testing and Yield Improvement

- **After fabrication**, chips undergo wafer test, package test, and final test.
- **Manufacturing defects are unavoidable.** Therefore testing is required to identify defective devices.
- **Failure mode analysis** (FMA) and **burn-in testing** improve yield and product reliability.

Design Verification Flow (1/3)



■ FIGURE 1.3

Design Verification Flow (2/3)

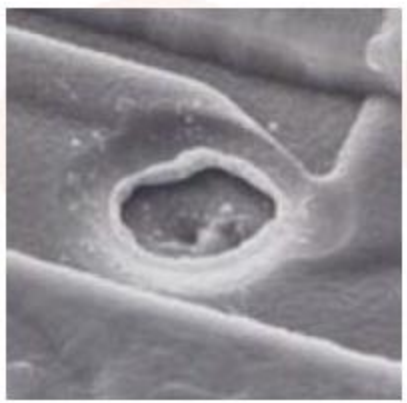
- **Behavioral → RTL:** Verify functional correctness.
- **RTL → Gate-Level:** Verify synthesized logic matches RTL behavior.
- **Gate-Level → Physical:** Verify timing and performance before fabrication.

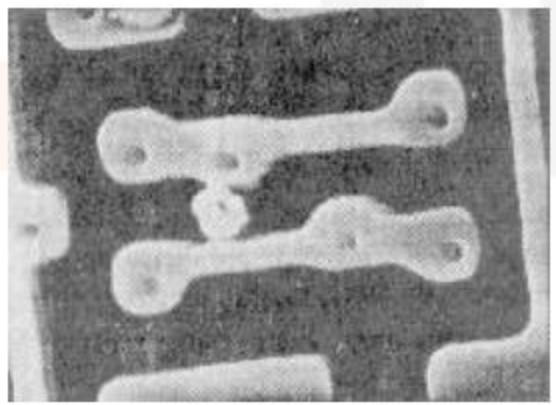
Design Verification Flow (3/3)

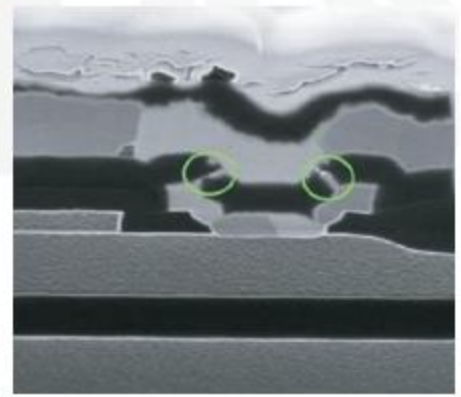
- Design verification relies on CAD tools, simulation, hardware emulation, and formal verification, but it remains a **time-consuming process**.
- Insufficient verification may fail to detect design errors, causing costly redesigns and delaying **time-to-market**.
- Verification techniques and test stimuli are often reused for **manufacturing testing**, linking design verification closely with test engineering.

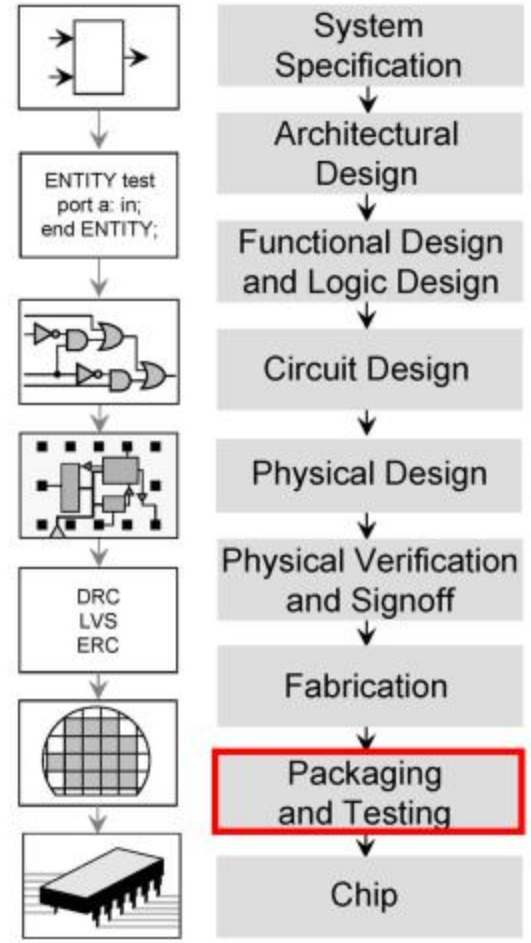
Production Testing

- Testing after chip fabrication in order to detect manufacturing defects that may impact functionality or electrical parameters.
- Manufacturing test ideally would check every node in the circuit to prove it is operational.



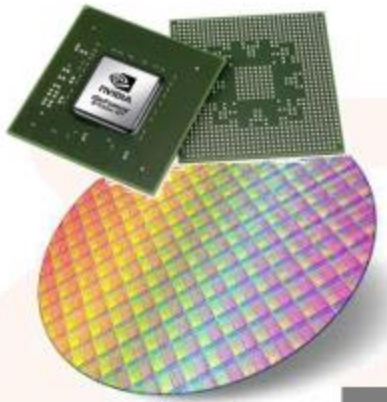






© Adam Teman, 2020

Production Test Flow



Test Patterns





Ship to Customers

Pass

Fail

Diagnostics and Yield learning



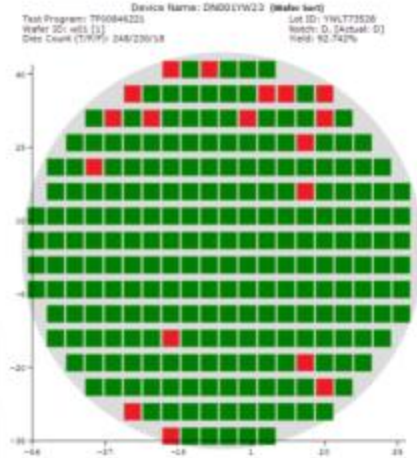
Yield Improvement

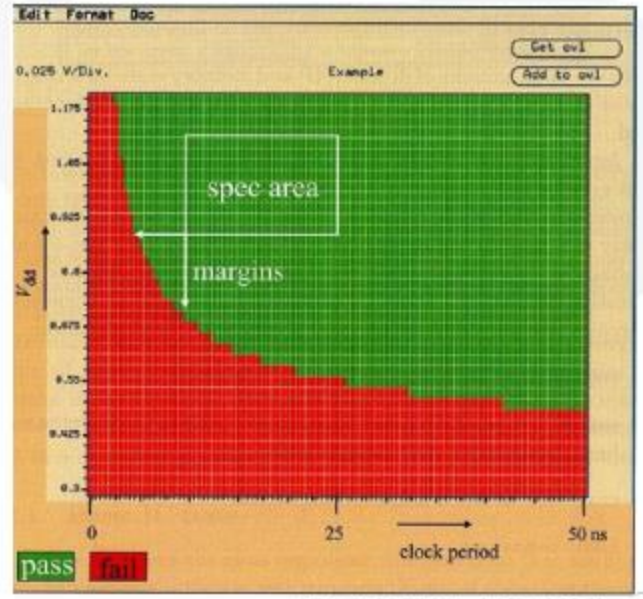
© Adam Teman, 2020

Production Test Flow

- Wafer level testing (Wafer Sort)
- Assembly & Packaging
- Open/Short test
- Packaged device test
- Burn-In (@ elevated voltage and temperature)
- Final Test (pass/fail) and Bin Sorting

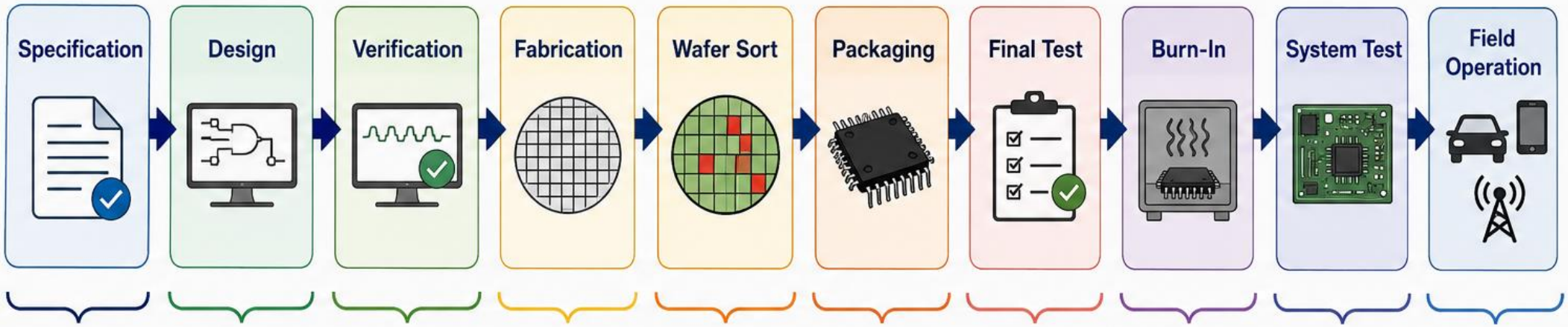
- Parametric Tests (voltage, temperature and clock)
  - Shmoo plot





© Adam Teman, 2020

Testing Throughout the VLSI Lifecycle



Specification	Design	Verification	Fabrication	Wafer Sort	Packaging	Final Test	Burn-In	System Test	Field Operation
Requirements definition	Design rule check, simulation	Functional verification, logic debug	Process defects may be introduced	Detect bad dies before packaging	Assembly process may cause defects	Detect defects at device level	Screen early-life failures	Validate in the target system	Ensure reliability in real world

Key Points

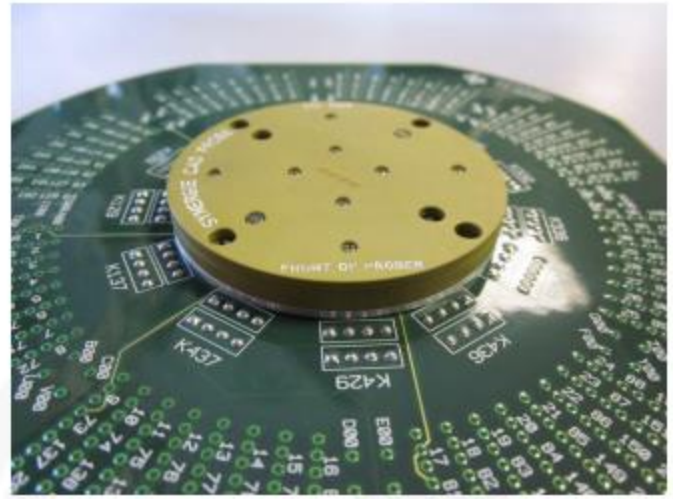
- Testing occurs at multiple stages in the VLSI lifecycle.
- Different tests target different failure mechanisms.
- DFT (Design for Testability) must be considered during design phase.

Key Message

Testing is not a single step — it is a complete lifecycle activity.

Wafer Sort

- **Wafer sort or probe test**
  - Done before wafer is *scribed* and cut into chips
  - Includes test site characterization – specific test devices are checked with specific patterns to measure gate threshold, poly sheet resistance, etc.
- **Probe card**
  - Custom built PCB to allow performing wafer sort
  - Modern probe cards can test an entire 12" wafer with one touchdown
  - Can contact several dies in parallel (~1-16)
  - Camera in the wafer prober allows alignment



Source: Synergie-CAD



Source: Anysilicon

© Adam Teman, 2020

Yield and Reject Rate

- The yield of a manufacturing process is defined as the percentage of acceptable parts among all parts that are fabricated.

$$Yield = \frac{Number\ of\ acceptable\ parts}{Total\ number\ of\ parts\ fabricated}$$

Yield and Reject Rate

- These two outcomes are often due to a poorly designed test or the lack of design for testability (DFT).

$$Reject\ rate = \frac{Number\ of\ faulty\ parts\ passing\ final\ test}{Total\ number\ of\ parts\ passing\ final\ test}$$

Yield and Reject Rate

- The ratio of field-rejected parts to all parts passing quality assurance testing is referred to as the reject rate, also called the defect level.
- The goal of six sigma manufacturing, also referred to as zero defects, is 3.4 PPM or less.

Four Possible Outcomes

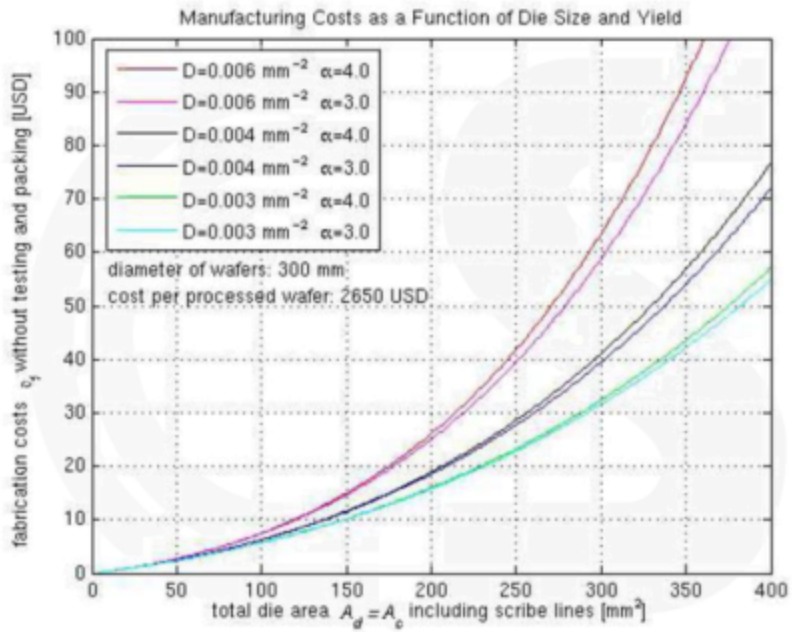
- True pass and true reject are correct decision
- *Test escapes* = defective chips that pass test
  - also known as (aka.) *under-testing*
- *Yield loss* = good chips that fail the tests
  - aka. *overkill*, *over-testing*
- Goal of good testing: reduce both test escape and yield loss
  - Trade off between test cost and test quality
    - Quality test reduces test escape but increases yield loss
    - Low cost test reduces yield loss but increase test escape

	Good IC	Defective IC
Pass tests	True PASS	Test Escapes (less is better)
Fail tests	Yield Loss (less is better)	True Reject

VLSI Test 1.1 © National Taiwan University

Yield & Cost

- **Die Yield** $= \frac{\#\ good\ dies}{\#\ manufactured\ dies} \approx \left(1 + \frac{D \cdot A}{\alpha}\right)^{-\alpha}$
  - D – Defect Density – average number of defects per unit of chip area
  - A – Chip area, α – Clustering Factor
  - ex. AxD=1, α=0.5 → Y=0.58
- $Die\ Cost = \frac{Wafer\ Cost}{Dies\ per\ Wafer * Die\ Yield}$
- $IC\ Cost = \frac{Die\ Cost + Testing\ Cost + Packaging\ Cost}{Final\ Test\ Yield}$
- **Defect Level (DL)**
  - The ratio of faulty chips among the chips that pass test, measured in DPM (<<500 DPM)



$$DL = 1 - Y^{(1-FC)},$$
$$0 < DL < 1 - Y$$

© Adam Teman, 2020

Yield

- According to the **Poisson** model, $Y = e^{-AD}$, the chip yield Y decreases exponentially with increasing area A and manufacturing defect density D.

Bài 1 — Tính Die Yield cơ bản

Một chip được chế tạo với:

- mật độ lỗi công nghệ là $0.004\ defects/mm^2$
- diện tích chip là $100\ mm^2$
- hệ số clustering bằng 2

Hãy tính yield của chip.

$$\boxed{Y \approx 69.4\%}$$

Bài 2 — Yield theo mô hình Poisson

Một IC có:

- diện tích chip $150\ mm^2$
- defect density bằng $0.005\ defects/mm^2$

Hãy tính yield của IC theo mô hình Poisson.

$$Y \approx 47.2\%$$

Bài 3 — Tính số good dies

Một wafer silicon tạo ra được 800 dies.

Biết rằng yield của wafer là 82%.

Hỏi có bao nhiêu good dies trên wafer?

$$656\ good\ dies$$

Bài 4 — Tính Die Cost

Một nhà máy semiconductor có:

- wafer cost bằng 3000 USD
- số dies trên wafer là 600
- die yield bằng 75%

Hãy tính die cost.

$$\boxed{6.67\ USD}$$

Bài 5 — Tính IC Cost

Một IC có:

- die cost bằng 7 USD
- testing cost bằng 1.5 USD
- packaging cost bằng 2 USD
- final test yield bằng 90%

Hãy tính IC cost của sản phẩm.

$$\boxed{11.67\ USD}$$

Bài 6 — Tính Defect Level

Một chip có:

- yield bằng 95%
- fault coverage bằng 99.5%

Hãy tính defect level của chip.

256 DPM

Bài 7 — So sánh chip nhỏ và chip lớn

Hai chip được sản xuất bằng cùng một công nghệ với defect density bằng $0.004\ defects/mm^2$.

- Chip A có diện tích $80\ mm^2$
- Chip B có diện tích $200\ mm^2$

Chip	Yield
A	72.6%
B	44.9%

- Hãy:
  1. Tính yield của từng chip.
  2. So sánh kết quả.
  3. Giải thích vì sao chip lớn thường có manufacturing cost cao hơn.

Bài 8 — Ảnh hưởng của Fault Coverage

Một chip có yield bằng 95%.

Hãy tính defect level trong hai trường hợp:

- fault coverage bằng 90%
- fault coverage bằng 99%

Sau đó:

1. so sánh kết quả
2. giải thích vai trò của DFT trong việc giảm DPM.

FC	DL
90%	5100 DPM
99%	510 DPM

Bài 9 — Bài toán tổng hợp Yield & Cost

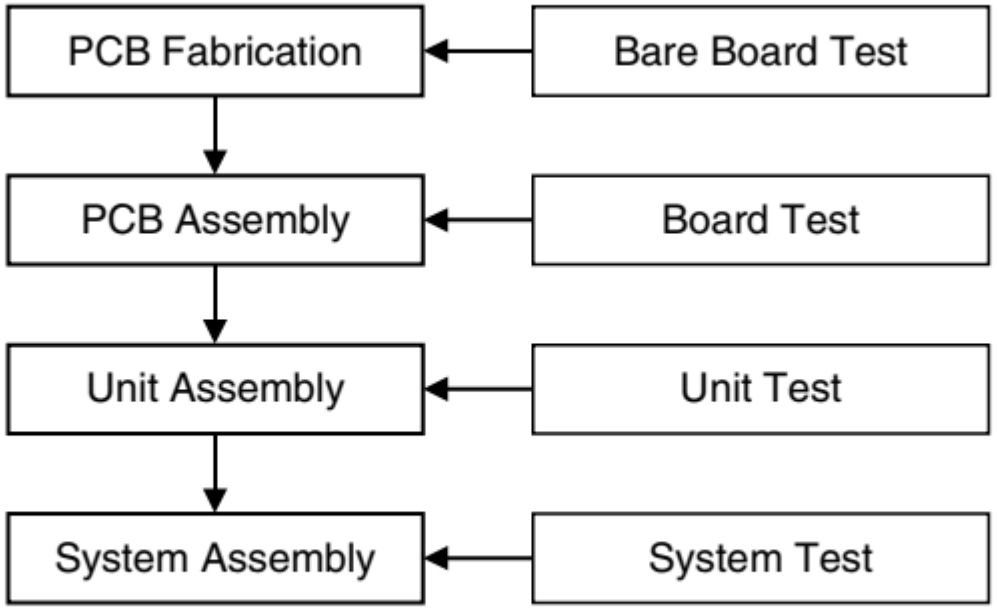
Một wafer có các thông số: wafer cost bằng 4000 USD , số dies trên wafer là 500, defect density bằng $0.003\ defects/mm^2$, chip area bằng $120\ mm^2$, clustering factor bằng 3. Ngoài ra: testing cost bằng 1 USD, packaging cost bằng 1.5 USD, final test yield bằng 95%

Hãy:

1. Tính die yield.
2. Tính die cost.
3. Tính IC cost.
4. Nhận xét ảnh hưởng của defect density và chip area đến manufacturing cost.

Đại lượng	Kết quả
Die Yield	71.2%
Die Cost	11.24 USD
IC Cost	14.46 USD

Electronic System Manufacturing Process (1/2)



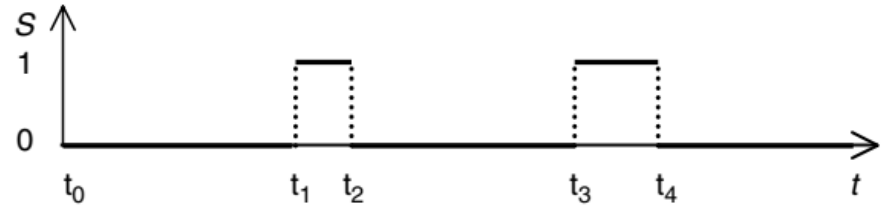
■ **FIGURE 1.4**

Electronic System Manufacturing Process (2/2)

- The steps required to manufacture an electronic system are also susceptible to defects.
- As a result, testing is required at these various stages to verify that the final product is fault-free.

System-Level Operation and Reliability

- During system operation, a number of events can result in a system failure
- Suppose the state of system operation is represented as S, where $S = 0$ means the system operates normally and $S = 1$ represents a system failure.



■ FIGURE 1.5